UNIVERSITY OF BAHRAIN
COLLEGE OF BUSINESS ADMINISTRATION
DEPARTMENT OF MANAGEMENT AND MARKETING
QM458 - ARTIFICIAL INTELLIGENCE

REINFORCEMENT LEARNING FOR OPTIMIZING MARKETING DISCOUNT STRATEGIES

DONE BY:

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SECTION: 01

INSTRUCTOR: DR. MOURAD MESSAADIA

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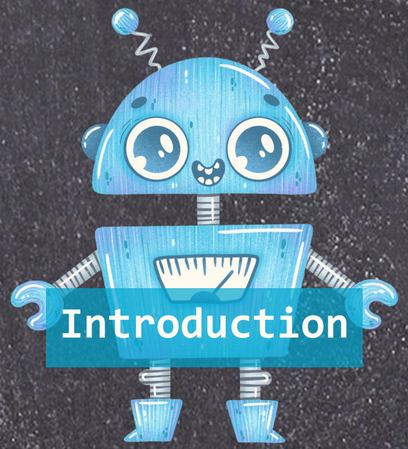
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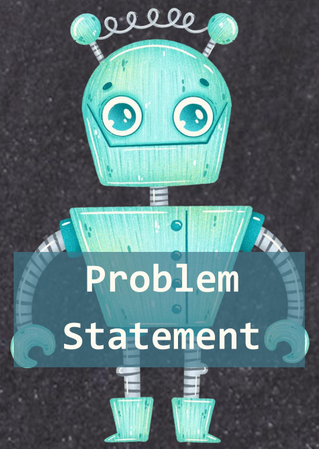
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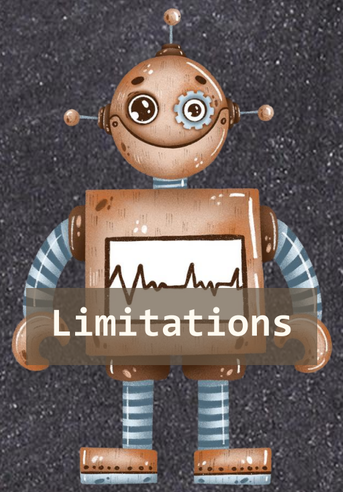
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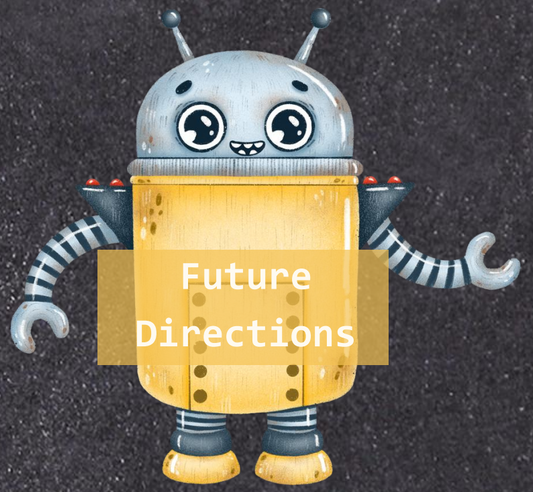
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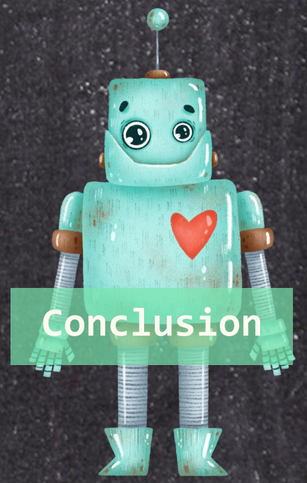
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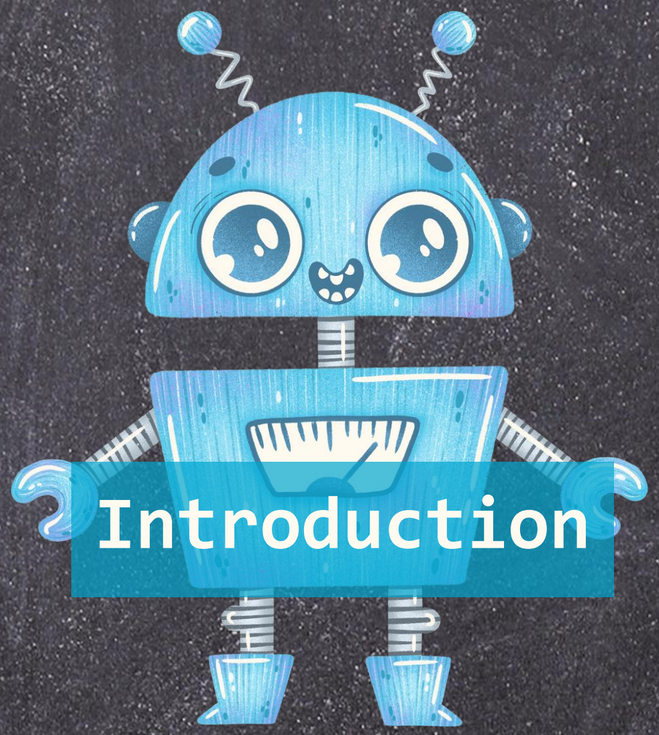

Introduction

In this presentation, we will explain how reinforcement learning can help businesses offer the right discount to different types of customers.

We focused on customer income levels by dividing them into low, medium, and high-income groups. Then, we tested different discount options using Python, such as no discount, 10%, 20%, and special offers during events.

Our goal is to help businesses improve sales by making smarter discount decisions without reducing overall profits.

< LITERATURE REVIEW

According to the study by Dayan and Niv (2008), RL works like the human brain – it tries something, sees if the result is positive or negative, and learns from it.

There are two types of models in RL:
- Model-free: learns directly from experience.
- Model-based: learns from planning ahead.

The study showed that RL is useful for making smart decisions in different situations, like offering the right discount to customers based on their status.

>

< LITERATURE REVIEW

Loftus et al. (2020) explained how reinforcement learning can help in making fast medical decisions under pressure, just like doctors do in emergencies.

The agent looks at different patient cases in their study and chooses the best medical action. If the decision is good, the agent gets a reward. If it's wrong, the agent gets a penalty. Over time, the agent learns from each case and gets better and faster at making decisions.

This shows that reinforcement learning is great at making smart choices very quickly, even in sensitive areas like healthcare.

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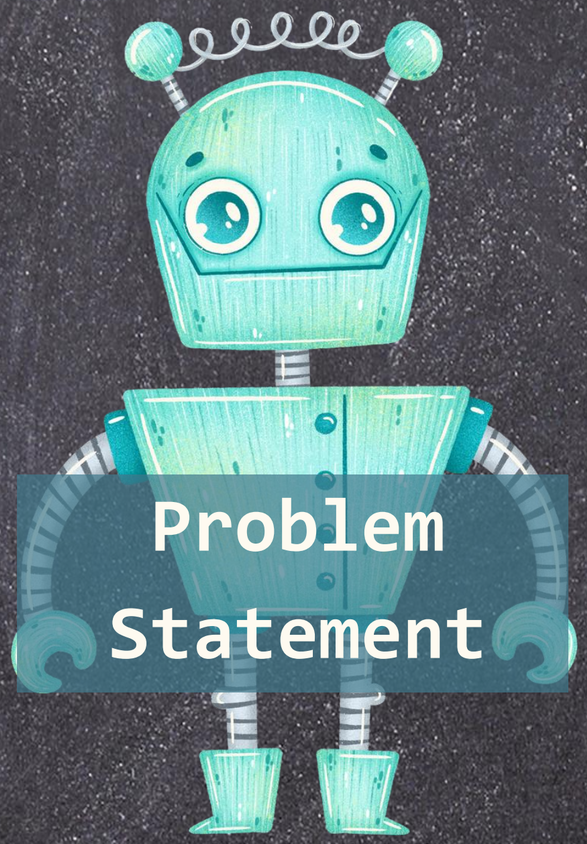

< Problem

Companies often struggle to choose the right discount for different customers.
Giving the same discount to everyone can waste money or miss sales.

For example:
- A high discount to wealthy customers reduces profit.
- A small discount to low-income customers may not be enough.

There’s a need for a smart system that learns which discount works best for each customer type based on their income level.

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< Approach
We used Reinforcement Learning (RL), specifically the Q-
learning algorithm.

The agent interacts with different income groups:
- Low
- Medium
- High

And tests four discount options:
- No discount
- 10%
- 20%
- Special 30% (for holiydays)

The agent gets a reward when a purchase happens,
and learns over time which discount works best for each
group.

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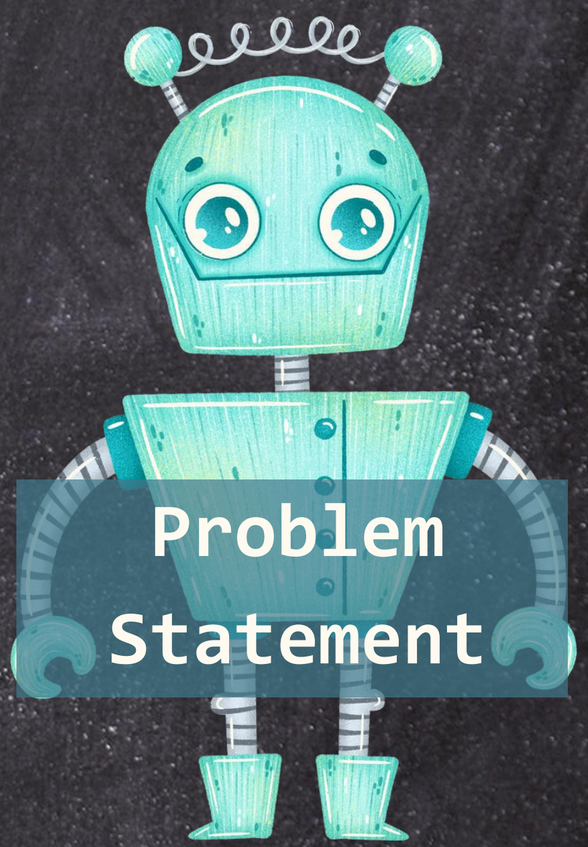
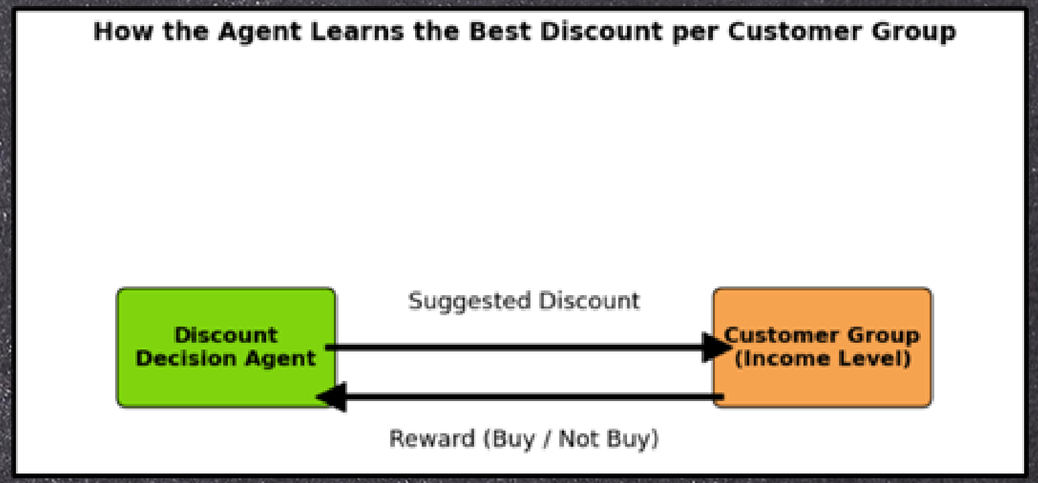
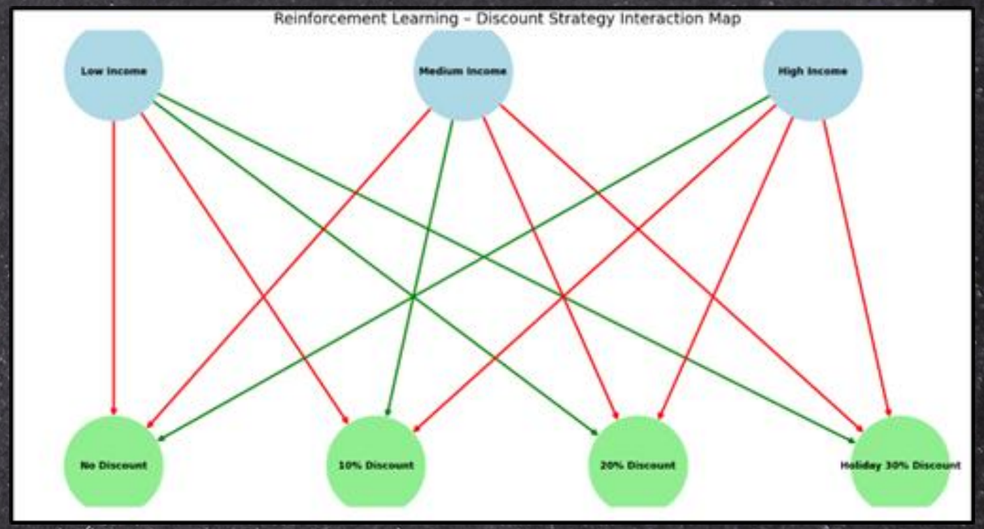


< RL Process >

The diagram below (Figure 1) illustrates how the agent improves its decisions over time through this learning cycle.





< RL Discount Strategy – Interaction Map >
Figure 2: This map shows how the agent interacts with different income levels and discount options.
>

```
Q-table after training:
[[3.18097698 1.79070354 5.38545398 3.33504732]
 [4.57553201 3.20681031 1.82904329 2.89847049]
 [5.60972436 2.42844858 2.81643892 3.14179078]]
```

< Figure 3: Q-table after training showing learned values for each discount per income level.
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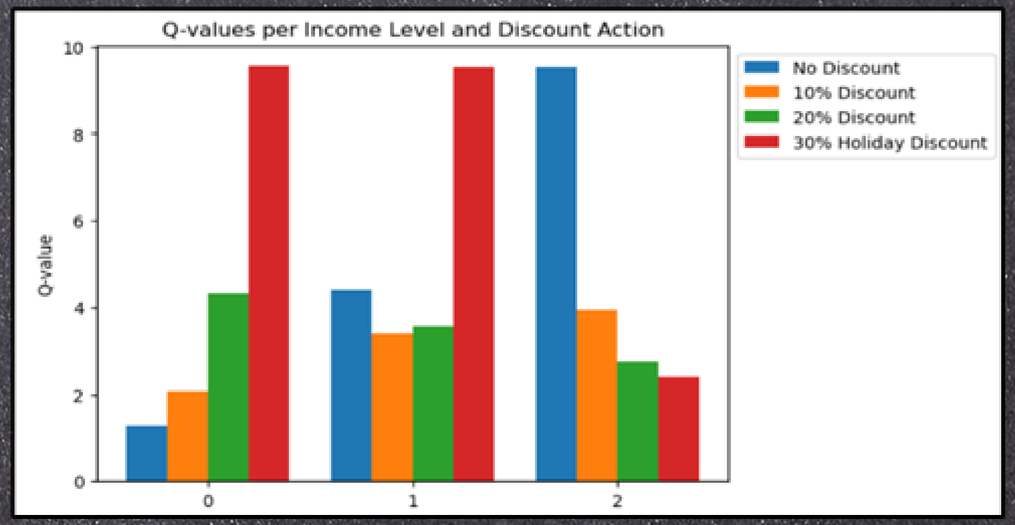

< Figure 5: Grouped Bar Chart – Q-values per Income Level and Discount Action

This chart visualizes the Q-values for each discount strategy across income levels. It helps compare which discount performs best for each group based on the agent's learning.

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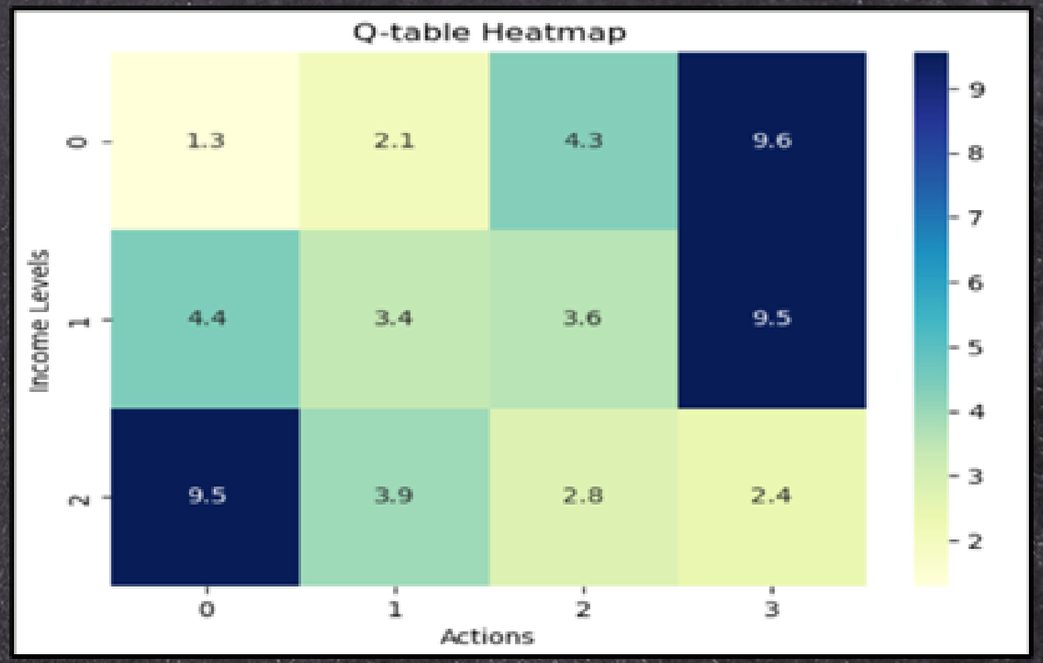

< The heatmap (figure 6) shows the best discount for each income level. Darker colors mean better results. For low income, the holiday discount works best. For high income, no discount is more effective.

>

< Limitations

- The model only includes three income levels and four discount options, while real-world customer behavior is more complex.
- The agent is trained on simulated data rather than real customer data, which may reduce the accuracy of the results.
- The rewards are based on assumptions and may not reflect actual business profits or customer decisions.
- Important factors such as seasonal trends, customer preferences, and marketing efforts are not considered.
- The model involves randomness, so results can vary with each run unless a random seed is fixed.

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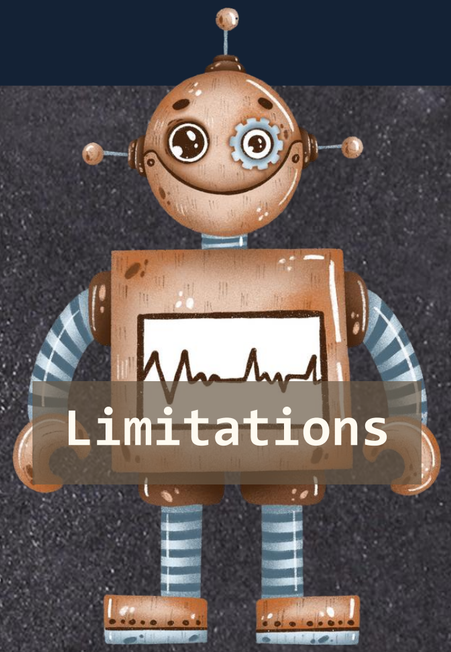

< Future Directions

Use Deep Q-Learning: To handle lots of data and make decisions faster.

Add new types of discounts: Like loyalty cards or gifts with products.

Consider customer behavior: How often they visit, how long they take to buy, and their preferences.

Offer seasonal discounts: That match holidays and special times.

Use customer satisfaction as feedback: Not just whether they bought, but how they felt about the offers.

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< Future Directions

Connect the system to real websites: Like Shein, to give suitable discounts.

Create a dashboard for marketing: To help the team monitor and adjust discounts easily.

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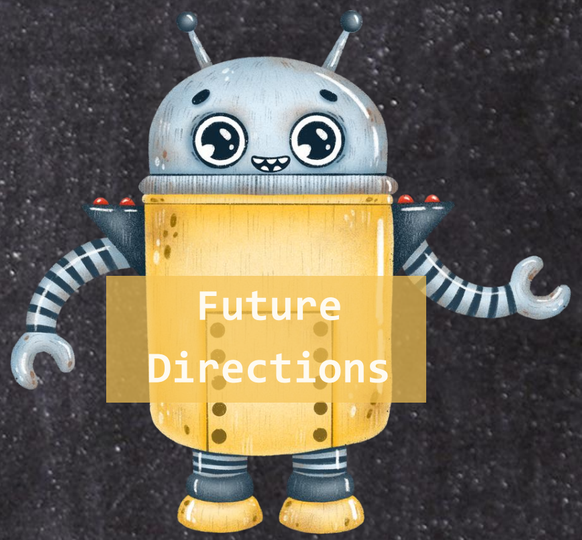
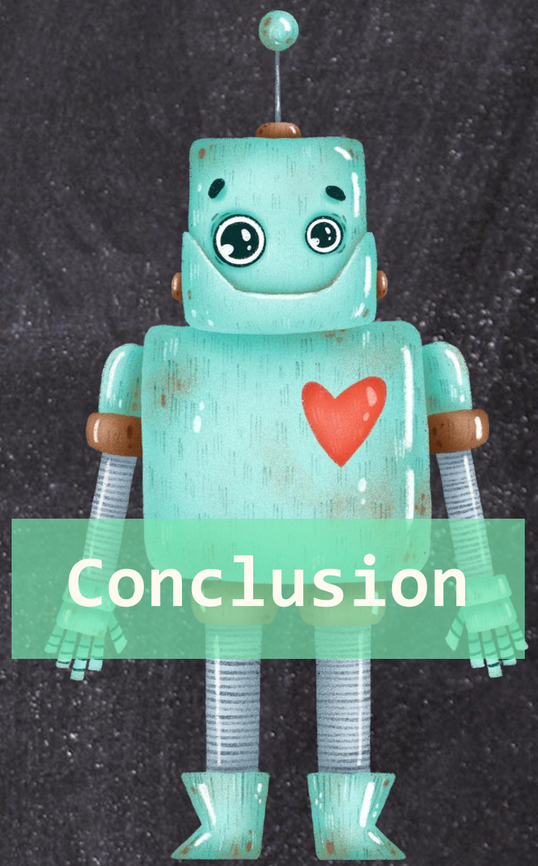


This project showed how reinforcement learning, especially Q-learning, can improve marketing decisions by giving each customer a discount based on their income.

The agent learns from customer responses and gets rewards to improve its decisions over time. It performed better than traditional marketing methods by making personalized choices that increase customer satisfaction and profits.

The agent is designed to be practical, flexible, and scalable for real-world use.



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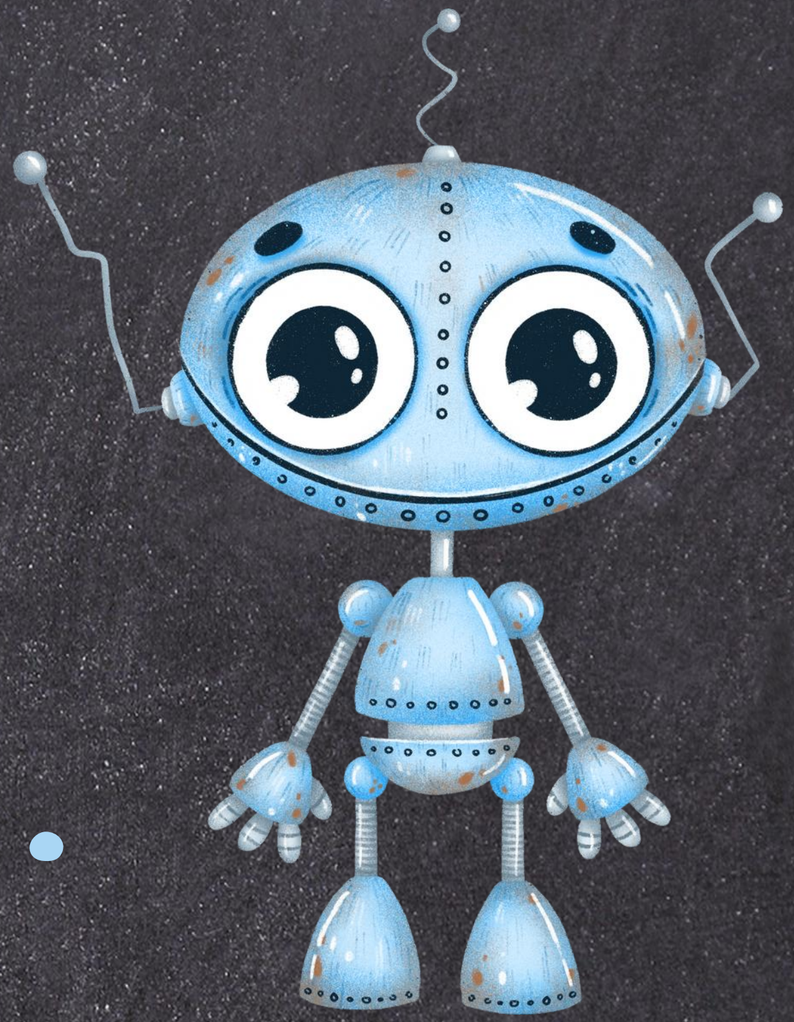
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